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## BEECH RANGE EXTENSION AND VEGETATION HISTORY: POLLEN STRATIGRAPHY OF TWO WISCONSIN LAKES<sup>1</sup>

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**Abstract.** The pollen stratigraphy of two small lakes in eastern Wisconsin (Radtke Lake, Washington County, and Gass Lake, Manitowoc County) records the Holocene (past 10 000 yr) spread of beech (*Fagus grandifolia*: Fagaceae). Radiocarbon dates were obtained for the oldest stratigraphic levels at which beech pollen appeared consistently in amounts >0.5% of terrestrial pollen.

A spatially continuous pattern of beech expansion from the north was ruled out, because beech trees grew in Wisconsin by 6000 BP, 2000 yr before adjacent populations were established to the north. Alternative geographic patterns of spread (from the south or east) were spatially discontinuous, requiring seed dispersal distances of perhaps 25–130 km. That beechnuts could be dispersed across such distances suggests (1) the involvement of Blue Jays, Passenger Pigeons, or other vertebrates, and (2) a capacity for reaching climatically controlled range limits, given sufficient time, despite such discontinuities in habitat. A lag of 1000–2000 yr between the establishment of source populations in Michigan and Indiana and the appearance of beech in Wisconsin suggests that low-probability dispersal events were involved and that dispersal constraints limited the range of beech during this time, although climatic and edaphic explanations for the lag cannot be ruled out.

Pollen data from the two sites reveal other features of vegetation history in eastern Wisconsin: an open *Picea*–*Fraxinus* woodland prior to 11 000 BP; a sequence of *Picea*, *Abies*, *Betula*, and then *Pinus* forests between 11 000 and 7500 BP; the establishment of a coniferous/deciduous forest ecotone (“tension zone”) ≈7000 BP in this region; and the presence of *Quercus*-dominated deciduous forests from 7000 BP until 110 BP (time of Euro-American settlement), a period punctuated by a gradual decrease in *Ulmus* populations (≈4500 and 5700 BP at the two sites) and by an increase in mesophytic tree abundance at the expense of *Quercus* after 3500 BP.

**Key words:** beech; biogeography; *Ectopistes migratorius*; *Fagus grandifolia*; migration; paleoecology; Passenger Pigeon; population expansion; range extension; seed dispersal; Wisconsin.

### INTRODUCTION

Dramatic changes have occurred in ranges of temperate-region forest trees since ≈17 000 yr ago, when the Wisconsin ice sheet began to melt (Denton and Hughes 1981). In response to atmospheric warming and the availability of newly deglaciated terrain, ranges of various species shifted northward at different rates and in distinctive geographic patterns in Europe and North America (Firbas 1949, Davis 1976, 1981, 1983, Bernabo and Webb 1977, Delcourt et al. 1983, Huntley and Birks 1983, MacDonald 1984, Bennett 1985).

One pattern of species range extension rather widely envisioned for postglacial tree migrations is diffusion (Pielou 1979), the gradual spread of a population across hospitable terrain. Gleason (1923) assumed that species ranges expanded in continuous, unbroken fronts along river valleys or other favorable pathways. Benninghoff (1964) also advocated this continuous model of range

extension, noting that expansive prairies of Indiana and Ohio acted as barriers to the migration of American beech and of eastern hemlock into Michigan from the south (Table 1 lists scientific names of plants in text; nomenclature follows Gleason and Cronquist 1963).

An alternative model for species range extension is jump dispersal (Pielou 1979), the spatially discontinuous and rather sudden spread of a population across unfavorable habitat where it cannot grow. For plants, jump dispersal involves long-distance seed dispersal and the establishment of outlier populations.

I used fossil pollen data from sediments of two small lakes to trace the Holocene (past 10 000 yr) migration of beech into Wisconsin. I probed changes over time in the beech range limit at discontinuities in favorable habitat, barriers to continuous spread through diffusion. In Wisconsin the presettlement range of beech was a band 30–100 km wide along the west shore of Lake Michigan (Fig. 1). The range of beech was continuous northeastward from Wisconsin into Upper Michigan but discontinuous in other directions, bounded on the east by Lake Michigan and on the south

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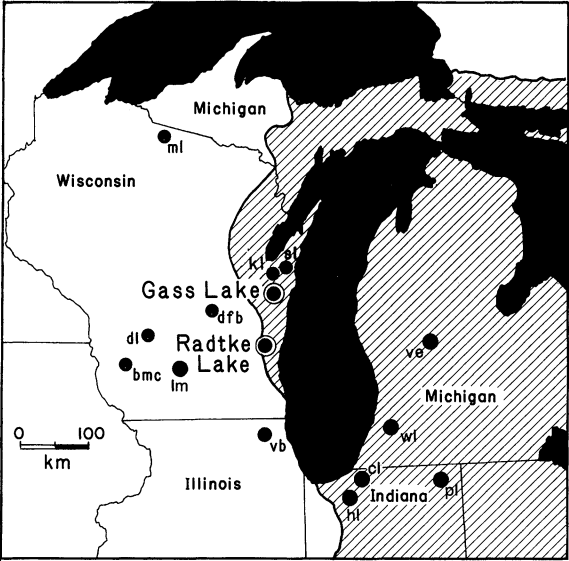


FIG. 1. The range of beech ( $\approx$ AD 1830, ▨), after Finley (1951) and Little (1971), and the location of study sites: Radtke Lake and Gass Lake, Wisconsin. Other workers' sites: bmc = Blue Mounds Creek, Wisconsin (Davis 1977); cl = Clear Lake, Indiana (Bailey 1972); dl = Devils Lake, Wisconsin (Maher 1982); dfb = Disterhaft's Farm Bog, Wisconsin (West 1961); hl = Hudson Lake, Indiana (Bailey 1972); kl = Kellners Lake, Wisconsin (Goodwin 1976); lm = Lake Mendota, Wisconsin (Winkler et al. 1986); ml = Mary Lake, Wisconsin (Webb 1974); pl = Pretty Lake, Indiana (Williams 1974); sl = Seidel Lake, Wisconsin (West 1961); ve = Vestaburg Bog, Michigan (Gilliam and Kapp 1967); vb = Volo Bog, Illinois (King 1981); wl = Wintergreen Lake, Michigan (Manny et al. 1978).

by oak savannahs and prairies of the xeric Prairie Peninsula in Illinois (Fig. 1; Transeau 1935).

Beech could have spread into Wisconsin through diffusion only from the north, through Upper Michigan, a migration route tentatively suggested by Kapp (1977) on the basis of Benninghoff's (1964) model of the Prairie Peninsula as a filter barrier to plant migration. Diffusion from the north would be in evidence if beech pollen data showed a time-transgressive pattern of beech arrival dates from north to south.

Alternative migration patterns require jump dispersal either 90–130 km from the east across Lake Michigan, as suggested by Curtis (1959) and Kapp (1977), or shorter distances from the south between scattered mesic sites within the Prairie Peninsula. The direction and timing of beech spread into Wisconsin is revealed by comparing beech pollen data from my two Wisconsin sites (Radtke Lake, Washington County; Gass Lake, Manitowoc County) with data from other sites in the Upper Midwestern United States (Gilliam and Kapp 1967, Bailey 1972, Kerfoot 1974, Williams 1974, Lawrenz 1975, Manny et al. 1978, King 1981, Futyma 1982, Davis et al. 1986, and K. D. Woods and M. B. Davis, *personal communication*).

These results also address a related question: what is the capacity of beech, a large-seeded tree that spread

slowly through North America during the Holocene (Davis 1976, 1981), for long-distance seed dispersal? In order to understand past or present range limits we must draw the difficult distinction between constraints on seed dispersal and constraints on establishment. Some recent range expansions that followed long periods of stasis seem to result from climatic change and concomitant release from constraints on establishment (Jacobson 1979, Grimm 1983, 1984). In other instances, relative stability of range limits might result from constraints on seed dispersal, particularly if barriers are larger than average distances of seed dispersal (Benninghoff 1964). Climatic and dispersal constraints operate in concert, although they often exert influence at different spatial and temporal scales (Davis et al. 1986, T. Webb 1986).

Little is known about actual patterns and distances of seed dispersal (Howe and Smallwood 1982), despite an abundance of anecdotal literature on seed dispersal mechanisms (Ridley 1930, Pijl 1982) and despite investigations of seed rain within small local study areas (Werner 1975, Rabinowitz and Rapp 1980, Lacey 1981, Reichman 1984). The probabilities of seed dispersal over distances  $>1$  km have been investigated through studies of seed-predator behavior (Bossema 1979, Darley-Hill and Johnson 1981, Johnson and Adkisson 1985). Paleoecological data can also help elucidate patterns and maximum distances of seed dispersal, by tracing shifts in species ranges over time. Information about dispersal distances is needed not only for interpretation of Holocene range extensions but also for estimating interpopulation gene flow and for monitoring population dynamics.

The pollen data presented here also document the Holocene vegetation history of eastern Wisconsin. No

TABLE 1. Scientific names of plants mentioned in text. Nomenclature follows Gleason and Cronquist (1963).

| Common name          | Scientific name           |
|----------------------|---------------------------|
| American beech       | <i>Fagus grandifolia</i>  |
| Ash                  | <i>Fraxinus</i> spp.      |
| Balsam fir           | <i>Abies balsamea</i>     |
| Basswood             | <i>Tilia americana</i>    |
| Black ash            | <i>Fraxinus nigra</i>     |
| Bur oak              | <i>Quercus macrocarpa</i> |
| Butternut            | <i>Juglans cinerea</i>    |
| Eastern hemlock      | <i>Tsuga canadensis</i>   |
| Elm                  | <i>Ulmus</i> spp.         |
| Hickory              | <i>Carya</i> spp.         |
| Hornbeam             | <i>Carpinus</i> spp.      |
| Ironwood             | <i>Ostrya virginiana</i>  |
| Jack pine            | <i>Pinus banksiana</i>    |
| Larch                | <i>Larix laricina</i>     |
| Oak                  | <i>Quercus</i> spp.       |
| Paper birch          | <i>Betula papyrifera</i>  |
| Ragweed              | <i>Ambrosia</i> spp.      |
| Red pine             | <i>Pinus resinosa</i>     |
| Spruce               | <i>Picea</i> spp.         |
| Sugar maple          | <i>Acer saccharum</i>     |
| White pine (eastern) | <i>Pinus strobus</i>      |
| Yellow birch         | <i>Betula lutea</i>       |

previous pollen sites with Holocene radiocarbon dates were situated in the southeastern Wisconsin vegetation zone that includes beech but excludes pine and hemlock. Other fossil pollen data for this zone come from Seidel Lake, Kewaunee County (West 1961), whose pollen stratigraphy is not dated except at the base, and from Kellners Lake, Manitowoc County (Goodwin 1976), for which unpublished radiocarbon age determinations for the Holocene stratigraphy have been made recently (M. B. Davis, *personal communication*; Fig. 1). Other southern Wisconsin sites of pollen studies with Holocene radiocarbon dates are located west of the range of beech (Fig. 1; West 1961, Webb 1974, Davis 1977, Maher 1982, Winkler 1985, Winkler et al. 1986, and R. G. Baker, *personal communication*).

#### BEECH AUTECOLOGY

American beech (*Fagus grandifolia*: Fagaceae) is a mesophytic, shade-tolerant, deciduous tree that does not tolerate fire, drought, or flooding (Fowells 1965). It grows in abundance in mesic hardwood forests throughout eastern North America. Beech regenerates by root sprouts and with seeds (Ward 1961, Held 1983). The propagules of beech are nuts 1–2 cm in diameter (hereafter “beechnuts” or “seeds”) that are dispersed by seed predators, such as squirrels, Blue Jays, and, prior to their extinction, Passenger Pigeons (Ridley 1930, Firbas 1949, Johnson and Adkisson 1985, S. Webb 1986). Beech was among the last tree taxa to reach the northern midwest (Davis 1976, 1981). Its relatively slow migration rate might have resulted from biotic constraints, such as dispersal limitations or its relatively late age for first reproduction, or from environmental constraints, such as scarcity of sites where edaphic and climatic conditions were favorable for establishment.

#### STUDY SITES

Two small lakes in southeastern Wisconsin were selected for their location with respect to the range of beech and for their small size, closed drainage, and accessibility. Both sites are south of the limit of Great-lakean Substage Glaciation 11 800 yr ago, situated in areas that were deglaciated by  $\approx 12\,500$  yr ago (Acomb et al. 1982, McCartney and Mickelson 1982) but were not inundated by the later Nipissing high-water stage of Lake Michigan.

Radtke Lake (cored 17 September 1980) is located in Washington County (Sections 22 and 27, Township 11 N, Range 20 E),  $\approx 70$  km from the southernmost extent of and at the western edge of the range of beech in Wisconsin (Fig. 1). At its center this 4-ha lake is 3.8 m deep and is underlain by 8.3 m of sediment dating back to 11 290 BP (radiocarbon years before present). Radtke Lake lies at an elevation of 900 m in sandy soils adjacent to an esker, within what was once deciduous forest of beech, sugar maple, basswood, and a mixture of oak species (Finley 1976). At the time of

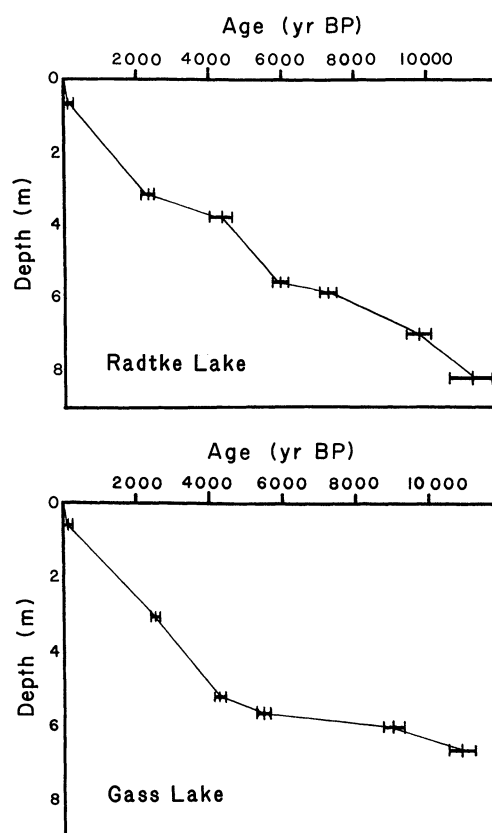


FIG. 2. Corrected radiocarbon age vs. depth in sediment cores: Radtke Lake and Gass Lake. Crosses represent radiocarbon-dated samples. Line width is radiocarbon date counting error.

settlement, nearby forests contained beech trees (M. B. Davis et al., *personal communication*). Between AD 1840 and 1850, southeastern Wisconsin's Euro-American population expanded, and wheat farms replaced large tracts of forest. Today, uplands near Radtke Lake are cultivated, developed residentially, or planted with jack pine. Swamps of larch and eastern white cedar occupy adjacent lowlands.

Gass Lake (cored 28 May 1981) is located in Manitowoc County (Section 3, Township 18 N, Range 23 E), 80 km north of Radtke Lake and well within the range of beech at the time of Euro-American settlement (Fig. 1). Gass Lake occupies  $\approx 2.5$  ha. At its center the lake is 6.8 m deep and is underlain by 8.8 m of sediment. Sediments 7 m below the lake bottom were deposited 11 030 BP. Located on glacial till at an elevation of 700 m, Gass Lake was surrounded by presettlement forests similar to those near Radtke Lake (Finley 1976). Beech was extremely abundant nearby (M. B. Davis et al., *personal communication*). Regional logging began in 1835, and wheat farms were established nearby between AD 1840 and 1850 (Smith 1973). Gass Lake is now rimmed by farmland, pasture, and small remnants of deciduous forest.

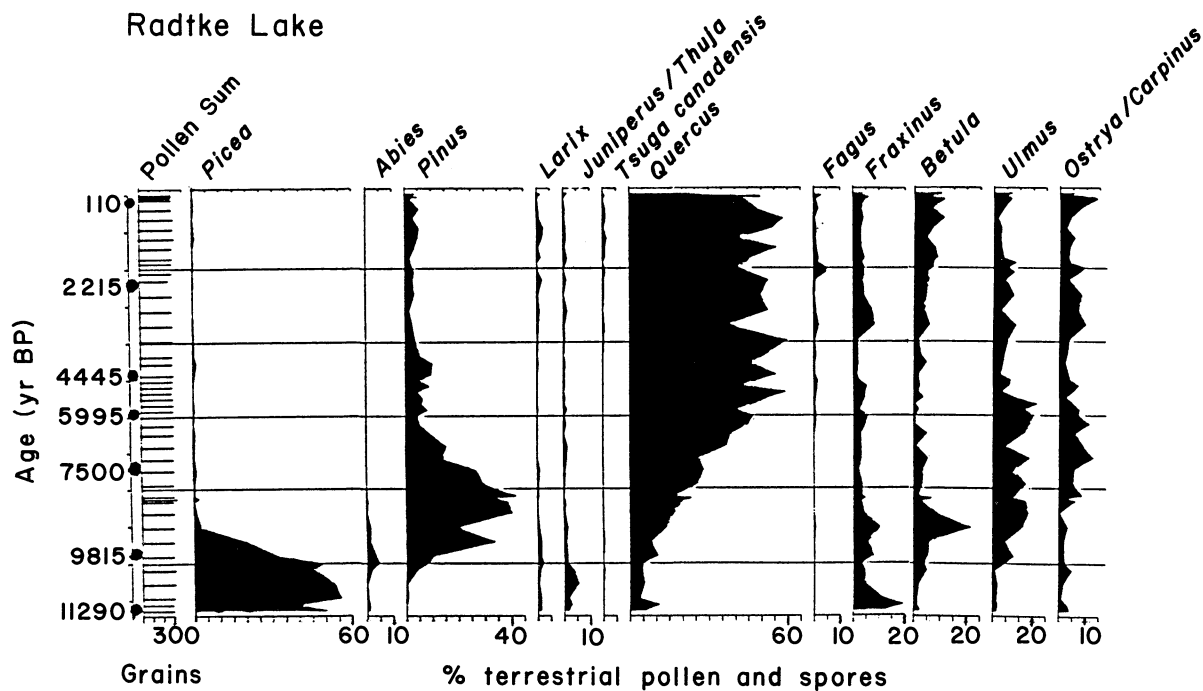


FIG. 3. Radtke Lake pollen diagram. Percent of total terrestrial pollen and spores vs. corrected radiocarbon age (yr BP = years before present) for major taxa.

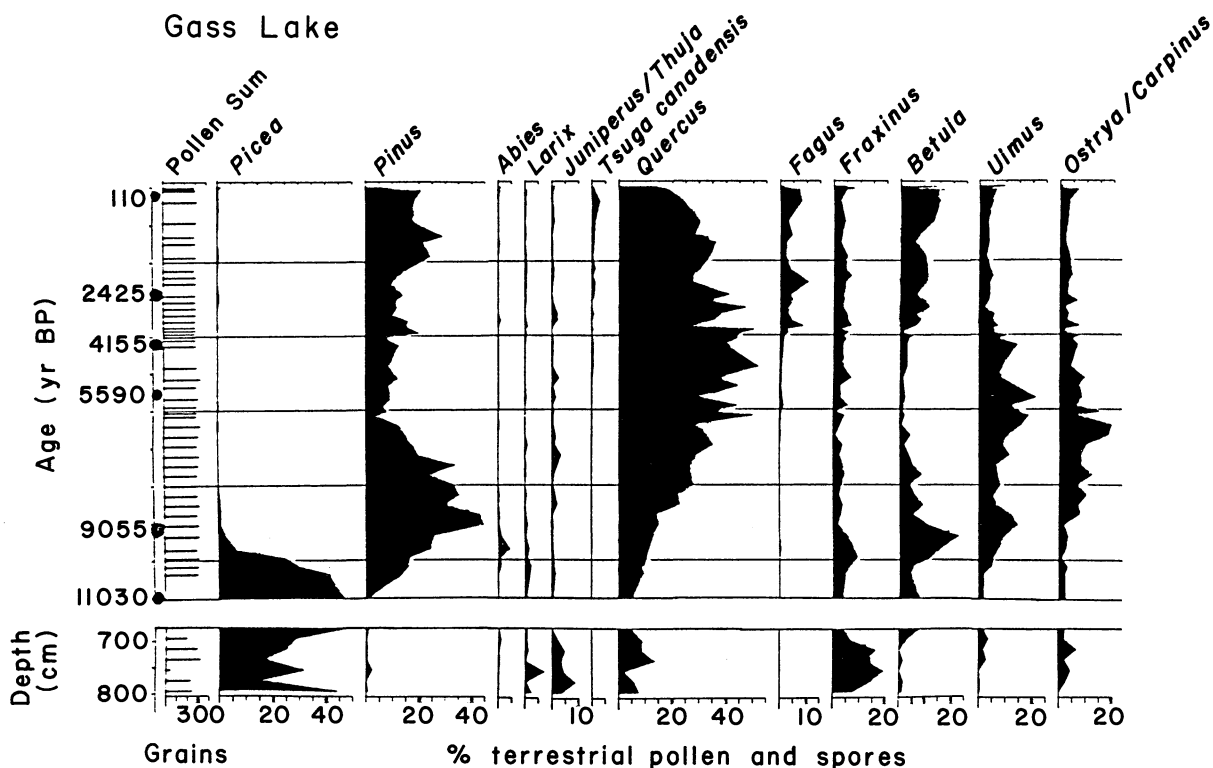


FIG. 4. Gass Lake pollen diagram. Percent of total terrestrial pollen and spores vs. corrected radiocarbon age (yr BP = years before present) for major taxa. For the oldest metre of the core, pollen percentages are plotted against depth rather than age.

Radtke Lake

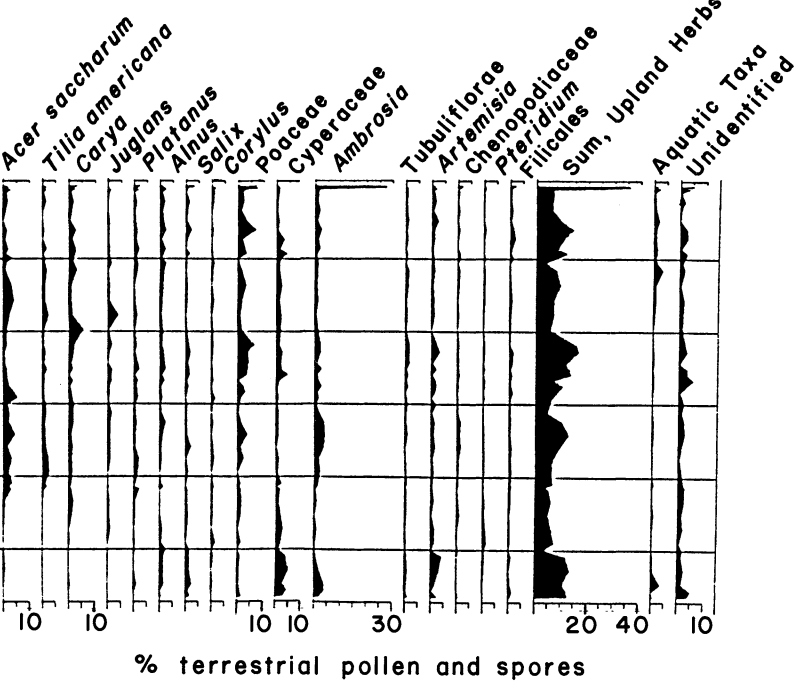


FIG. 3. Continued.

Gass Lake

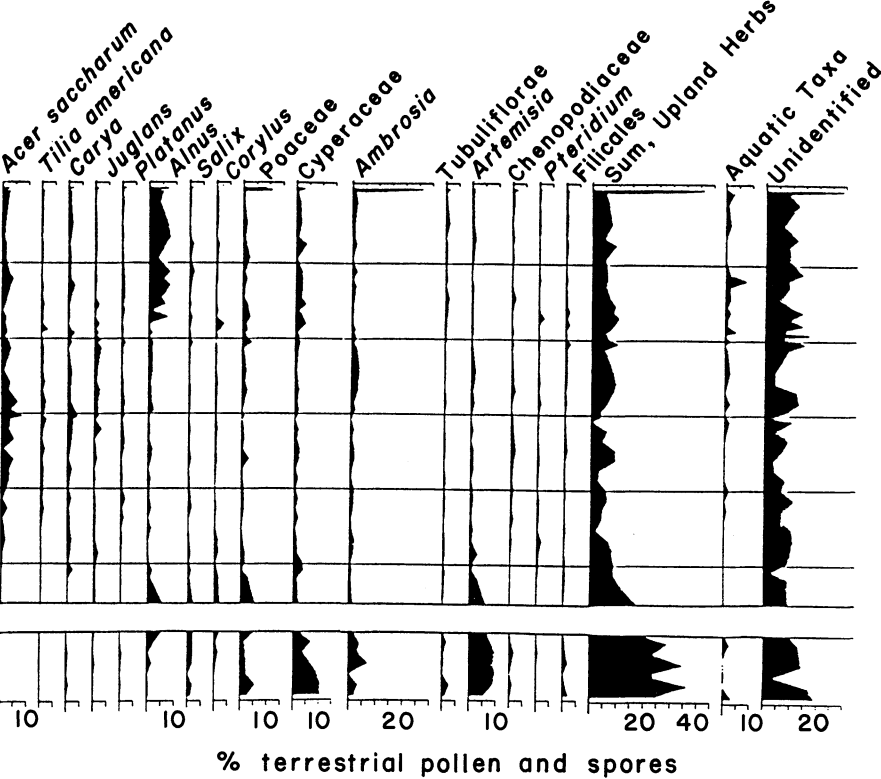


FIG. 4. Continued.



TABLE 2. Radiocarbon dates for sediment samples from Radtke Lake and Gass Lake, Wisconsin. Dates are in radiocarbon years before present (AD 1950). See Methods for explanation of date correction.

| Sample number | Depth in core (cm) | Uncorrected radiocarbon age (yr) | Corrected radiocarbon age (yr) |
|---------------|--------------------|----------------------------------|--------------------------------|
|               | Range (midpoint)   |                                  |                                |
| Radtke Lake   |                    |                                  |                                |
| GX7887        | 52–66 (59)         | 280 + 115                        | 110                            |
| GX7888        | 312–320 (316)      | 2385 + 145                       | 2215                           |
| GX7889        | 400–407 (403)      | 4615 + 180                       | 4445                           |
| GX7890        | 556–564 (560)      | 6165 + 185                       | 5995                           |
| GX7891        | 605–611 (608)      | 7670 + 185                       | 7500                           |
| GX7892        | 700–708 (704)      | 9985 + 260                       | 9815                           |
| GX7893        | 826–835 (830)      | 11 460 + 580                     | 11 290                         |
| Gass Lake     |                    |                                  |                                |
| GX8563        | 59–67 (63)         | 840 + 125                        | 110                            |
| GX8564        | 311–319 (315)      | 3155 + 165                       | 2425                           |
| GX8565        | 517–525 (521)      | 4885 + 145                       | 4155                           |
| Beta6269      | 545–553 (594)      | 6320 + 120                       | 5590                           |
| GX8566        | 610–618 (614)      | 9785 + 235                       | 9055                           |
| GX8567        | 669–681 (674)      | 11 760 + 380                     | 11 030                         |

METHODS

I used a modified Livingstone corer (Wright et al. 1965) to remove 6.3 cm diameter sediment cores in 1-m sections from the deepest point in each lake. Plastic tubes 7.5 cm in diameter were used to recover the mud-water interface and underlying flocculent, water-saturated sediment.

Sediment samples of 1 cm<sup>3</sup> were removed from these cores, combined with known quantities of exotic *Eucalyptus* pollen (Benninghoff 1962), and processed for pollen analysis according to procedures described by Faegri and Iversen (1975). Samples with high clay content were sieved using Nitex screening (mesh size 7 μm) to remove clay particles (Cwynar et al. 1979). For each sample, at least 300 terrestrial pollen grains were identified and tabulated along evenly spaced transects across microscope slides. Counts were made at 54 levels in the Gass Lake core and at 51 levels in the Radtke Lake core, at an average temporal interval of ≈200 yr.

Radiocarbon dates were obtained for sediment from seven levels in the Radtke Lake core and from six levels in the Gass Lake core (Table 2). High temperature ashing (Dean 1974) of Gass Lake sediments indicated that large amounts of carbon were present as CaCO<sub>3</sub> (5–8% of total dry mass), a sign that old carbon derived from carbonate bedrock might have been incorporated into sediments. Carbon derived from pre-Quaternary bedrock would not have been in isotopic equilibrium with atmospheric carbon when sediment was deposited; such nonequilibrium carbon can bias radiocarbon dates (Deevey et al. 1954). To check this possibility I dated the rise in ragweed pollen that marks settlement by Euro-Americans, which occurred in southeastern Wisconsin in AD 1840 (Smith 1973), 110 yr before the

AD 1950 radiocarbon base year. The Gass Lake ragweed pollen rise, at a depth of 63 cm, was dated at 840 ± 125 BP (years before present), confirming that older carbon was incorporated into sediments. The Radtke Lake ragweed pollen rise, at a depth of 59 cm, was dated at 280 ± 115 BP, indicating a smaller error.

To correct for nonequilibrium carbon I subtracted 730 yr from each radiocarbon date for Gass Lake and 170 yr from each date for Radtke Lake, making the assumption that the ratio of nonequilibrium carbon to equilibrium carbon has been constant throughout the lakes' histories (Deevey et al. 1954, Brubaker 1975). Radiocarbon dates for both lakes should be treated as approximations, but agreement of major pollen-stratigraphic events since 7000 yr ago suggests the error is no greater than perhaps 200–300 yr (Webb 1982). All dates reported here are based on corrected C<sup>14</sup> years.

Plots of age against depth for radiocarbon-dated samples (Fig. 2) show that sedimentation rates have fluctuated over time in both lake basins. Curve-fitting efforts were unsuccessful because even the best-fit transformations had high residual values in several portions of the pollen profiles, including the time of beech pollen appearance and increase. Hence a date was assigned to each level in the cores by linear interpolation between adjacent radiocarbon dates.

No radiocarbon dates were available for the lower 2 m of the Gass Lake core. Sediment below the 681-cm level was primarily clay with insufficient organic matter for reliable dating. The extrapolation of a constant rate of deposition to the core bottom leads to an unreasonably old bottom date of 16 500 yr, 4000 yr older than the date of deglaciation in the area (McCartney and Mickelson 1982). The basal clay zone probably represents a period of more rapid sediment deposition such as Kerfoot (1974) reported for Frains Lake, Michigan.

RESULTS AND DISCUSSION

*Pollen stratigraphy and vegetation history*

Pollen diagrams for Radtke and Gass Lakes trace the history of the vegetation of southeastern Wisconsin over the past 11 400 yr (Figs. 3 and 4). Pollen data and diagrams of pollen accumulation rates are found in Webb (1983). Major vegetation changes generally fit into regional patterns described by Webb et al. (1983) and were similar to those seen elsewhere in southern Wisconsin (Goodwin 1976, Maher 1982, Winkler et al. 1986).

*Spruce-ash woodland.*—The Gass Lake profile begins with a spruce-herb pollen assemblage containing high percentages of pollen of nonwoody, herbaceous plants, primarily *Artemisia*, other Asteraceae, and the Cyperaceae, along with moderately abundant (20–35%) spruce pollen, and abundant (>10%) black ash pollen. This combination suggests either an open spruce-black ash woodland with herbs in openings or a treeless vege-

tation. Analogous vegetation producing a similar pollen spectrum does not exist today (Wright 1971, Overpeck et al. 1985). In either case, vegetation near Gass Lake was more open before 11 000 BP than it was thereafter, when spruce pollen rose to the 50% level and herbaceous pollen dropped sharply, from 30 to 10%.

The absence of such a treeless or open woodland zone from the Radtke Lake pollen profile has two possible explanations: (1) that Gass Lake formed earlier, recording an open woodland phase also present further south near Radtke Lake but unrecorded there because of persistent ice in the lake basin; or (2) that these two sites just 80 km apart were surrounded by different vegetation types at the same time. In that case the situation may be analogous to that in Minnesota where herb-dominated tundra persisted until 10 000 yr ago in northeastern Minnesota while spruce forest covered other parts of the state (Watts 1967). Dated sediment cores from other area lakes would help to distinguish between these two possibilities.

*Spruce forests.*—By 11 000 BP, a relatively closed spruce forest covered southeastern Wisconsin. Like other early postglacial spruce forests of the midwest, this vegetation differed from modern North American spruce forests in its lack of pine and in the presence of deciduous trees, particularly black ash (West 1961, Cushing 1965, Wright 1971, Maher 1982, Webb et al. 1983). At both Radtke and Gass Lakes, ash pollen declined (from 20 to 5%) early in the spruce forest period, with spruce pollen percentages increasing at the apparent expense of black ash (Figs. 3 and 4).

Dominance of spruce declined throughout eastern North America between 12 000 and 9000 BP (Wright 1971, Bernabo and Webb 1977, Webb et al. 1983). In Wisconsin, percentages of spruce pollen dropped sharply  $\approx$ 10 000 yr ago (Figs. 3 and 4). Subsequently both sites had brief, distinct peaks (5%) in pollen of balsam fir, suggesting an abundance of balsam fir trees around 10 000 BP. This brief balsam fir pollen peak was followed by a brief birch (probably paper birch) pollen peak (20%)  $\approx$ 9000 BP.

*Pine forests.*—As spruce abundance declined  $\approx$ 10 000 BP, pines became more numerous. In eastern Wisconsin, pine dominated pollen spectra and vegetation from 10 000 to 8000 BP. Diploxylon pines (jack pine or red pine or both) were present by 10 000 BP or earlier, and white pine, the region's only native Haploxylon pine species, was apparently present by 9000 BP and predominated thereafter. Maher (1982) also found an early (9600 BP) arrival of white pine in southern Wisconsin, in contrast with its delayed appearance further north and west (Jacobson 1979, Webb et al. 1983).

*Deciduous forests.*—Beginning  $\approx$ 10 000 BP, a number of deciduous trees increased in importance, particularly elm, oak, and ironwood or hornbeam. Other woody angiosperms first appeared in eastern Wisconsin between 10 500 and 8000 BP, in order: butternut,

hickory, sugar maple, and basswood. Appearing much later were beech and (apparently) yellow birch (Figs. 3 and 4).

Pine gradually declined in importance between 7000 and 8000 BP, while oak and ironwood/hornbeam (pollen indistinguishable) increased in abundance. Deciduous forest persisted in southeastern Wisconsin from  $\approx$ 7000 until 100 BP, changing from time to time in details of composition and dominance. Throughout this period, important trees near both Radtke and Gass lakes included oak, sugar maple, basswood, ash, elm, yellow birch, ironwood or hornbeam, and, after it arrived in the region, beech.

Both pollen diagrams indicate a mid-Holocene decrease in elm population size  $\approx$ 5700 BP near southerly Radtke Lake and 4500 BP near Gass Lake (Figs. 3 and 4). Percentages of elm pollen dropped abruptly (from 15 to 5%) and remained low thereafter. A similar decline in elm abundance is indicated at other midwestern sites (Webb et al. 1983) but often with less abruptness. It occurred at various times: 5500 BP at Devils Lake, Sauk County (Maher 1982), 5200 BP at Blue Mounds Creek, Richland County (Davis 1977), 4000 BP at Frains Lake, southeastern Michigan (Kerfoot 1974), and 4600 BP at Volo Bog, northeastern Illinois (King 1981). This decrease in abundance of elm could have been caused by pathogenic outbreaks, competitive interactions, climatic or edaphic change, or combinations of these factors. Of special interest is the general synchronicity of the elm decline and the arrival of beech near Gass and Radtke lakes. However, beech was never present at many sites that nonetheless exhibit declines in elm (Davis 1977, King 1981, Maher 1982, Winkler et al. 1986).

Mid-Holocene pollen spectra from Gass and Radtke lakes fail to elucidate the timing of a climatic warming trend documented at some midwestern pollen sites. Conditions became warmer and dryer between 8000 and 6000 BP in Minnesota and northern Wisconsin (Bartlein et al. 1984). At other sites farther south in Wisconsin and in Illinois, similar changes were of lesser magnitude (Bartlein et al. 1984) and later in occurrence (after 6500 BP: Winkler et al. 1986). The relative insensitivity of my southeastern Wisconsin sites to such a change suggests that no ecotonal shifts occurred in the vicinity. The forests present when beech arrived  $\approx$ 6000 BP were mesophytic hardwoods stands where oak was perhaps more abundant than previously and where elm, yellow birch, and possibly sugar maple were somewhat less abundant than previously.

A later return to cooler, more moist conditions after 3500 BP (Winkler et al. 1986) is clearly in evidence in southeastern Wisconsin. At both Radtke and Gass lakes, oak pollen declined, while pollen of pine, birch, sugar maple, and beech increased in abundance after 4000 BP. Percentages of alder (*Alnus*) pollen suddenly increased in Gass Lake sediment  $\approx$ 3500 BP and remained high ever since, perhaps indicating a lake level



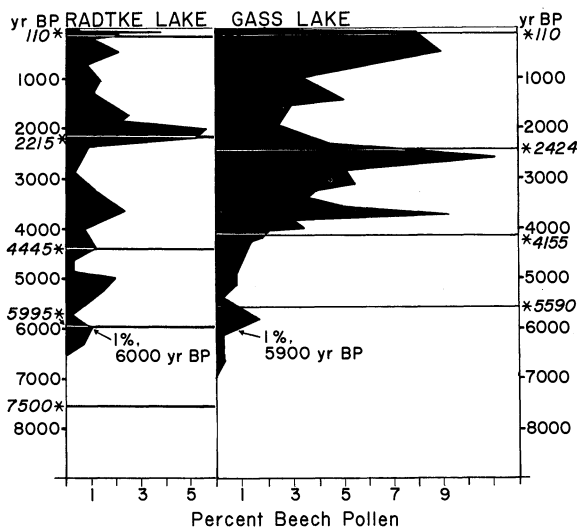


FIG. 5. Beech pollen percentage (of total terrestrial pollen and spores) vs. corrected radiocarbon age (yr BP = years before present), Radtke Lake and Gass Lake, Wisconsin. Horizontal lines indicate radiocarbon-dated levels (\* = radiocarbon date).

change like that at Lake Mendota where alder pollen also increased in abundance around that time (Winkler et al. 1986).

Several differences between the pollen diagrams from Radtke and Gass lakes correspond with a north-south climatic gradient but also could have resulted in part from edaphic differences between the two sites. Better-drained soils of the esker adjacent to near Radtke Lake might have supported drought- or fire-tolerant herbs absent near Gass Lake, where mesophytic plants might have been favored. However, between-site differences in pollen spectra were not great and generally corresponded with site proximity to presettlement vegetation regions as described by Curtis (1959) and Finley (1976). These pollen differences did not emerge until  $\approx 7000$  BP, suggesting that the ecotone between Wisconsin's northerly coniferous-deciduous forests and southerly deciduous forests (the "tension zone" of Curtis 1959) first became established near its present location after that time (see also Webb et al. 1983).

Both lakes are in the deciduous forest region south of Curtis' tension zone, but at the time of settlement Gass Lake was only 20 km from mixed coniferous-deciduous forests. Radtke Lake, 80 km further south, was closer to extensive oak openings and prairies of south-central Wisconsin (Finley 1951, 1976). Reflecting this difference in location, pine pollen percentages were consistently higher at Gass Lake (10–20%) than at Radtke Lake (3–10%) after 7000 BP. Just another 50 km north of Gass Lake, Seidel Lake sediments had still greater pine pollen percentages (20–40%: West 1961), although sediments from Kellners Lake, 22 km north of Gass Lake, had less pine pollen ( $\approx 5\%$ : Goodwin 1976). Throughout the Holocene, herbaceous pol-

len, especially that from grasses, was slightly more abundant at more southerly Radtke Lake (6–15%) than at Gass Lake (4–8%), although closed forest grew around both lakes. Sediments in areas of open oak woodlands have much more herbaceous pollen, as at Blue Mounds Creek, Wisconsin (Fig. 1; Davis 1977), where grass and ragweed pollen alone account for 40% of pollen in sediments throughout the middle and late Holocene.

Another difference between the Radtke Lake and Gass Lake pollen diagrams involves beech, which since 4000 BP contributed 2–10% of all pollen in sediments of Gass Lake but which only rarely contributed  $>2\%$  of pollen in sediments of Radtke Lake. No such difference is seen in pollen contributions of other fire-intolerant, mesophytic deciduous trees such as sugar maple or basswood, and no corresponding difference is seen in pollen contributions by potential competitors. Beech pollen percentages similar to those at Gass Lake occur also at Kellners Lake (Goodwin 1976), 22 km to the north. Perhaps beech trees were less abundant at Radtke Lake throughout the Holocene, as they were just before settlement (M. B. Davis, *personal communication*). Radtke Lake is close to the southern and western edges of the Wisconsin range of beech, where fewer sites may have met the physical requirements of beech.

A century ago, at the time of settlement, tree pollen percentages decreased and herbaceous pollen percentages increased, including the pollen of grasses and of ragweed, a wind-pollinated agricultural weed.

### Beech in Wisconsin

The Gass Lake and Radtke Lake pollen records demonstrate that beech grew on the Wisconsin side of Lake Michigan by 6000 BP. An exact date could not be assigned to the arrival of beech populations near these lakes, because beech pollen did not increase suddenly from absence to abundance. Beech pollen reached the 1% level 6000 BP at Radtke Lake and 5900 BP at Gass Lake and was consistently present in sediments thereafter (Fig. 5). A similar pattern occurs in sediments of Kellners Lake, 22 km north of Gass Lake, where beech pollen reached the 1% level without subsequent disappearance  $\approx 5850$  BP (Goodwin 1976; M. B. Davis, *personal communication*).

The difficulties of inferring species presence or absence from fossil pollen data have been discussed elsewhere (Davis 1963, Faegri 1966, Anderson 1970, Birks and Birks 1980, Webb 1982, Bennett 1985, Davis and Jacobson 1985, Davis et al. 1986). For beech, several recent studies in Wisconsin and nearby Michigan establish the quantitative relationship between beech pollen in sediments and proximity of beech trees. In presettlement sediments in Wisconsin and Michigan, small quantities of beech pollen (0.5–1% of terrestrial pollen) can be found up to 20 km from the limit of the beech range, but beech pollen levels are always  $<0.5\%$  at greater distances (Davis et al. 1986). This conclusion

is based upon comparisons of presettlement pollen spectra and presettlement vegetation data. That beech pollen was not widely dispersed in this region is also demonstrated by two studies of modern (postsettlement) pollen: vegetation relationships. Only at sites within the range of beech do modern beech pollen percentages reach the 1% level (Webb et al. 1981), and small lakes collect few beech pollen grains from populations >4.5 km distant (Bradshaw and Webb 1985).

The small quantities of beech pollen (0.5–1.5%) that occurred in Wisconsin 6000–4000 BP can only have had their source in local populations, on the western shore of Lake Michigan, because beech forests on the eastern shore of Lake Michigan were 100–120 km distant. Hence beech was present on the Wisconsin side of the lake by  $\approx$ 6000 BP, although populations might have been small or located up to 20 km from Radtke and Gass lakes.

#### *Beech migration routes*

Beech apparently reached southeastern Wisconsin at least 2000 yr before it reached Upper Michigan, a difference greater than possible errors in the radiocarbon dates ( $\approx$ 200–300 yr: Webb 1982). The appearance of beech near Radtke and Gass lakes  $\approx$ 6000 BP precedes arrival dates of 3000–4000 BP at sites in Upper Michigan and northeastern Wisconsin (Davis et al. 1986; K. D. Woods and M. B. Davis, *personal communication*).

Beech must have moved into Wisconsin across habitat where it does not grow now, in a discontinuous, jump-dispersal pattern. One possible route was from east to west across Lake Michigan. A seed source area certainly existed east of the lake. By  $\approx$ 7400 BP, beech was abundant in western Lower Michigan and northern Indiana, as indicated by pollen data from several sites (Fig. 1; Gilliam and Kapp 1967, Bailey 1972, Williams 1974, Lawrenz 1975, Manny et al. 1978).

Until  $\approx$ 8000 BP, Lake Michigan occupied a smaller area and so posed a seed-dispersal barrier narrower than 25 km in some parts of the lake basin (Prest 1970). However, by the time beech appeared in Wisconsin,  $\approx$ 6000 BP, the lake was back to at least its present extent ( $\approx$ 130 km across: Prest 1970), and between 5500 and 4500 BP a high-water stage (the Nipissing) made the lake a somewhat broader barrier ( $\approx$ 140 km wide) than it is today (Hough 1963, Lewis 1969, Prest 1970, Saarnisto 1975, Cowan 1978). Seed dispersal distances on the order of 90 km (a minimum likely lake width after 7500 BP) to 130 km (the lake's present minimum width near southern Wisconsin) would have been required for the range of beech to expand across Lake Michigan.

A second possible migration route was from the south, through the Prairie Peninsula, a projection of xerophytic, fire-tolerant vegetation extending into mesic deciduous forests of the east-central United States (Gleason 1923, Transeau 1935, Wright 1970). It seems

likely that scattered favorable sites provided "stepping stones" through the 120-km stretch of relatively xeric savannahs and prairies that separated Wisconsin beech populations from the nearest continuous beech populations to the south and southeast.

Before Euro-American settlement ( $\approx$ AD 1830), forests were discontinuous in southeastern Wisconsin and northeastern Illinois (Transeau 1935, Finley 1951, 1976, Meyer 1952, Goder 1956, Kilburn 1959). Xerophytic bur oak was the dominant tree in a mosaic of tallgrass prairie, oak savannah, and oak-hickory woodlands. Beech was not mentioned in notes of the United States Land Office surveyors for Kane County, northeastern Illinois (Kilburn 1959), nor was it mentioned for a 25-km stretch of southeastern-most Wisconsin, in Racine County (Goder 1956). Beech does grow today in parts of Lake and Cook counties in northeastern Illinois (Mohlenbrock 1975). This recent beech presence shows up in sediments from Volo Bog, Lake County, Illinois (Fig. 1; King 1981), which contain up to 2% beech pollen over the past 500 yr but which had only infrequent beech pollen previously. Even during this recent (last 500 yr) beech expansion, the tree apparently has been restricted to scattered mesic or fire-protected sites, such as discontinuous bluffs along Lake Michigan or islands within wetlands.

Data from Volo Bog and climatic reconstructions for the region suggest that the distribution of beech in the Prairie Peninsula was perhaps more restricted during the middle Holocene than it was just before settlement. At present the Prairie Peninsula coincides with an area of low snowfall and frequent, severe summer drought (Bryson 1966, Wright 1970). These climatic parameters have changed over time, but the region was hotter and dryer 6000 BP than it is today (Wright 1970, Bartlein et al. 1984, Winkler 1985, Winkler et al. 1986).

The normal response of beech populations to hot, dry conditions is a decline (Bartlein et al. 1986). Thus a large continuous beech population in northern Illinois was even less likely 6000 BP than immediately prior to settlement. A jump-dispersal pattern of seed dispersal between scattered sites was apparently necessary for beech to spread into Wisconsin from the south.

Although dry mid-Holocene conditions precluded a continuous beech forest in the Prairie Peninsula, it is possible that such conditions paradoxically generated scattered patches of favorable beech habitat by exposing organic soils of former wetlands. This scenario is consistent with pollen data from Volo Bog, Illinois (King 1981). Around 5600 BP, beech pollen reached the 1% level at Volo Bog, suggesting that an outlying population was established in Illinois at about the same time beech reached the vicinity of Radtke and Gass lakes.

Seed dispersal across intervening prairies or savannahs was perhaps more likely than seed dispersal across the wider expanse of Lake Michigan. Available evi-

dence does not rule out either a southerly or easterly migration route. In either case, a continuous migration front was not the mode of beech range extension into Wisconsin.

### *Migrational time lag*

A second result of interest was an apparent 1000–2000 yr lag between the presence of source populations of beech in Lower Michigan and Indiana 7000–8000 BP (Fig. 1; Gilliam and Kapp 1967, Bailey 1972, Williams 1974, Lawrenz 1975, Manny et al. 1978) and the arrival of beech on the other side of Lake Michigan (in Wisconsin)  $\approx$  6000 BP (see maps in Davis et al. 1986).

This lag might have resulted from infrequency of long-distance beechnut dispersal to favorable establishment sites. Immigration probabilities generally decrease as distances between source and target increase (MacArthur and Wilson 1967, Carlquist 1974). Hence in heterogeneous landscapes it is not surprising to find delays between the opening of new favorable sites, through climatic or edaphic changes, and the dispersal of propagules to those sites. The relatively short duration of the lag period in question suggests that even large-seeded plants eventually reach climatically controlled equilibria within the Great Lakes region, but that dispersal constraints can disrupt the tracking of climatic changes by species distributions over shorter (<2000 yr) time periods (Davis et al. 1986, T. Webb 1986).

Dispersal patterns may have interacted with climatic change to influence the spread of beech into Wisconsin. Although the mid-Holocene warm, dry period would not directly promote beech abundance, a concomitant lower water table might have exposed organic soils favorable for beech at scattered sites within the Prairie Peninsula, as mentioned above. With new islands of habitat available, distances between favorable sites would have decreased and the rate of spread would have increased. Climatic changes might instead have exerted other indirect effects on beech populations, for example by reducing populations of seed predators or fungal pathogens and so increasing the supply of viable seed. Theoretically, the larger the seed supply, the greater the average and maximal seed dispersal distances (Skellam 1951).

### *Mechanisms for discontinuous beech range extension*

The discontinuous “jump-dispersal” pattern by which the range of beech apparently expanded into Wisconsin requires seed dispersal across distances of perhaps 25–100 km. Beechnuts lack fleshy pulp to attract birds and lack the small size and the wings that facilitate dispersal by wind. The major vectors of beechnut dispersal are probably seed predators (birds, squirrels, bear, deer) that occasionally drop seeds or overlook seed caches (Ridley 1930, Firbas 1949, S. Webb 1986). Of these,

birds are perhaps the most likely to move beechnuts across water or treeless prairie.

In North America, potential avian dispersal agents for beechnuts included the Blue Jay (*Cyanocitta cristata*) and the extinct (since AD 1914) Passenger Pigeon (*Ectopistes migratorius*). Recent studies demonstrate that jays cache acorns and beechnuts after dispersing them distances of 2–4 km or more (Bossema 1979, Darley-Hill and Johnson 1981, Johnson and Adkisson 1985). Unrecovered acorns and beechnuts may germinate in such caches where they are in close contact with mineral soil and protected from desiccation. Maximum dispersal distances are probably greater than those detectable with modern behavioral studies.

The Passenger Pigeon was also likely to disperse beechnuts over distances required for jump-dispersal of beech into Wisconsin (S. Webb 1986). Pigeons do not cache seeds but could incidentally disperse seeds by disgorging undigested crop contents (Schorger 1955) or when killed by birds of prey, which leave pigeon crops untouched. Passenger Pigeon abundance, nomadic habits, and foraging behavior strongly suggest a role in occasional beechnut dispersal over long distances (S. Webb 1986), distances like those involved in the Holocene extension of the range of beech into Wisconsin. By moving beechnuts across Lake Michigan, across northeastern Illinois, or between scattered sites in the Illinois Prairie Peninsula, Blue Jays or Passenger Pigeons could account for the discontinuous spread of beech into Wisconsin.

### CONCLUSIONS

The pollen stratigraphy of sediments from Gass and Radtke lakes reveals a Holocene vegetation history similar in its major features to that of other midwestern sites. An open woodland or treeless vegetation was replaced by closed, spruce-dominated forest  $\approx$  11 000 BP. After brief periods of fir and then birch dominance, pine increased in importance  $\approx$  10 000 BP. Deciduous forests then gradually replaced pine forests and prevailed from 8000 BP until forest clearance by Euro-American settlers a century ago.

The composition of Holocene deciduous forests shifted from time to time. Notable features of pollen profiles of both lakes include (1) the apparent establishment nearby of the coniferous-deciduous forest ecotone (the “tension zone” of Curtis 1959)  $\approx$  7000 BP; (2) the arrival of beech trees  $\approx$  6000 BP; (3) a gradual, asynchronous elm decline 5700 and 4500 BP at the two sites; and (4) an increase in mesophytic tree dominance at the expense of more xerophytic oaks  $\approx$  4000–3500 BP.

For beech, a spatially discontinuous pattern of range extension, from the south or east, appears likely. Beech grew in Wisconsin 2000 yr before contiguous populations were established to the north. The mid-Holocene scarcity of favorable beech habitat to the south (in xeric savannahs of the Prairie Peninsula region) and



to the east (open waters of Lake Michigan) suggest a jump-dispersal pattern of spread that involved beechnut dispersal across unfavorable habitat. Seed-eating birds such as Passenger Pigeons or Blue Jays probably were responsible. A lag of 1000–2000 yr between the appearance of beech forests in Lower Michigan and in Wisconsin might have resulted from low dispersal probabilities over the distance and habitats involved.

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